FFT anomaly detection vs Z-score

Can I detect anomaly using z-score instead of FFT: **NO**

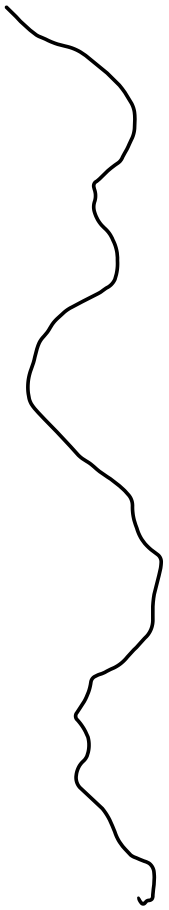
Last amended: 30th Oct 2026
#My folder: C:\Users\ashok\OneDrive\Documents\anomaly_detection\FFT\sensor_data\
Ref: Knime article here
Corresponding pdf file is also in the folder
GitHub place: Under Clustering and also hadoop repo
Virtual Machine: ubuntu_database under Documents/

Objective:

FFT essentially indicates anomaly in univariate data. So, even though Z-score may indicate no anomaly, FFT can indicate anomaly as the following experiment shows.

Our original data was pure sine wave (see below). For certain reasons, distortion came in. Can you say, just by calculating z-score from the data-values of the distorted sine wave that this data is anomalous. Z-score indicates no-anomaly in the distorted sine wave. But this distorted sine-wave is anomalous in the sense that it has more than one frequency, even though the amplitude of summed up distorted wave remains within bounds.

Data values	z_score
0	-0.008
0.866	1.479
-0.866	-1.495
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-0.866	-1.495
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0.524	0.892
0.5202	0.885
0.1852	0.31
0.2199	0.369
0.4006	0.68
-0.3198	-0.557
-0.4884	-0.847
0.1954	0.327
-0.9577	-1.653
-0.0946	-0.171
0.1429	0.237
-0.2281	-0.4
0.3754	0.637
0.559	0.952
0.2629	0.443
0.0391	0.059
0.6532	1.114
-0.6005	-1.039
-0.2281	-0.4
0.0005	-0.007
-0.8618	-1.488
-0.0747	-0.136
0.0108	0.01
-0.0066	-0.019

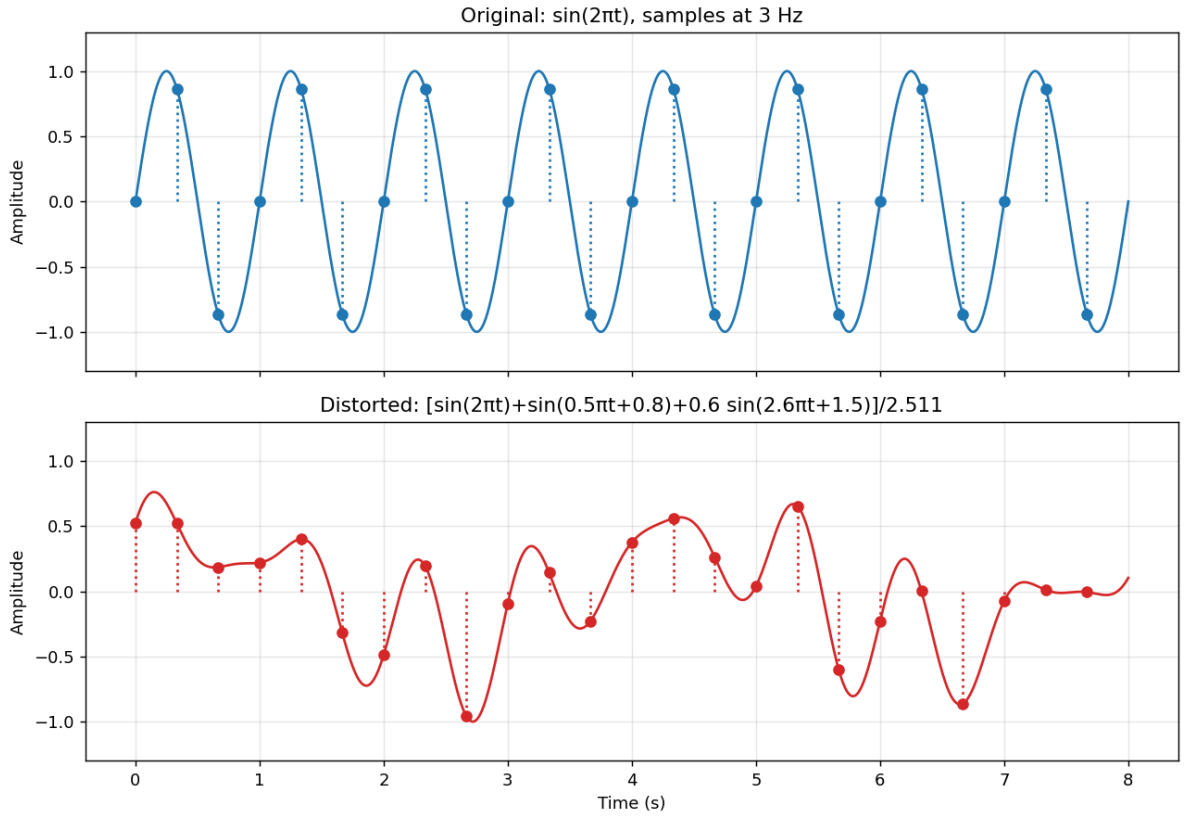


Figure 1: FFT would discover anomaly in the slower signal even though z-score would not.

Query:

Create a sine wave. Write down some of its points after sampling, say, three times its frequency. Introduce some distortion in this sine wave, keeping the frequency of one of its constituents the same as before but the frequency of other constituent(s) should be different. And also, the overall amplitude of the distorted wave remains the same as that of the earlier pure sine wave. Sample this sine wave also. And write down these sampled points.

Merge the first set of points and the second set of points to create a third set. Assume that this third set is normally distributed. Use z-score to find out anomalous points in this set of data. Please also plot both the original sine wave and the distorted sine wave.

Reply:

I made a version with much stronger distortion. The plots are in sine_waves_v2.png (original and distorted waves, with sample points) and overlay_v2.png (both waves overlaid).

1. Original wave

$x_1(t) = \sin(2\pi \cdot 1 \cdot t)$, sampled at 3 Hz for 8 s, which gives 24 points. The values are the same repeating pattern as before: 0, 0.866, −0.866, 0, 0.866, −0.866, ...

2. Strongly distorted wave

The distorted wave keeps the 1 Hz constituent and adds two others of different frequency, both with large amplitudes:

$x_2(t) = [\ \sin(2\pi t) + 1.0 \cdot \sin(0.5\pi t + 0.8) + 0.6 \cdot \sin(2.6\pi t + 1.5)\ ] / 2.511$

- The 1 Hz constituent is unchanged.
- The 0.25 Hz constituent has the same amplitude as the fundamental.
- The 1.3 Hz constituent has amplitude 0.6. It stays below the 1.5 Hz Nyquist limit at 3 Hz sampling, so it does not alias.

The divisor 2.511 is the true peak of the sum, found by searching over a long time span, so the peak amplitude of x_2 is exactly 1, the same as x_1. The waveform is no longer periodic over the 8 s window, so the pattern doesn't repeat from cycle to cycle.

Sampled points of x_2 (n = 0–23, t = n/3 s):

n	0–5	6–11	12–17	18–23
x_2	0.524, 0.520, 0.185, 0.220, 0.401, −0.320	−0.488, 0.195, −0.958, −0.095, 0.143, −0.228	0.375, 0.559, 0.263, 0.039, 0.653, −0.600	−0.228, 0.000, −0.862, −0.075, 0.011, −0.007

3. Merged set

The merged set has 48 points (24 from each wave). Its mean is ≈ 0.005 and its standard deviation is σ ≈ 0.582.

4. Z-score results

Group	z-score range
Original samples	0 and ±1.48 to ±1.50
Distorted samples	−1.65 to +1.11

The largest |z| is −1.65, at n = 32, where $x_2 = -0.958$.

- With the usual thresholds of |z| > 2 or |z| > 3, there are no anomalies.
- With a stricter threshold of |z| > 1.5, only that one point ($x_2 = -0.958$) would be flagged. The original wave's ±0.866 samples sit just under 1.5 (±1.48 to ±1.50).

Takeaway

The distortion is now clearly visible. The distorted wave no longer repeats and has irregular peaks and troughs. Z-scores still find almost nothing, because the amplitude constraint caps every value at ±1, so no point can be a statistical outlier.

If you want z-scores to catch the distorted points, you would have to drop the equal-amplitude constraint or inject spikes. A tool that compares the two waves' spectra would also show the distortion more clearly than z-scores do.

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